UK Road Safety Analysis: Accident Trends, Risk Factors Safety Insights

Data Visualization Analytics — DVA Capstone 2

Newton School of Technology, Sonipat

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Team ID	G-4 (Section E)
Project Title	UK Road Safety Analysis
Sector	Transportation & Public Safety Analytics
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Submission Date	April 29, 2026
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GitHub Repository	Click here
Tableau Dashboard	Dashboard Link

1. Executive Summary

Problem

Despite strict traffic regulations, the United Kingdom continues to experience a high number of road accidents and casualties each year, many of which are preventable. A major challenge for transport authorities and urban planners is the lack of clear, data-driven visibility into where and why severe accidents occur. Existing data is underutilized, making it difficult to identify high-risk locations and understand how factors such as weather, road conditions, lighting, and time influence accident severity. As a result, safety measures are often broad and generalized rather than targeted, reducing their overall effectiveness.

Approach

This project addresses the problem by analysing UK road accident data using a structured data pipeline. The dataset was sourced from publicly available government records and processed using Python for data cleaning, transformation, and exploratory analysis. Key variables such as accident severity, weather conditions, time of day, and road type were analysed to uncover patterns. The cleaned data was then visualized using an interactive Tableau dashboard, enabling stakeholders to explore trends and identify high-risk areas in a simple and intuitive manner.

Key Insights

- Accident severity is strongly influenced by environmental conditions, with poor weather and low visibility significantly increasing risk.
- Nighttime and low-light conditions show a disproportionately higher share of severe and fatal accidents.
- Accidents are concentrated during peak traffic hours, indicating increased risk due to congestion and high vehicle density.
- Certain road types and locations consistently report higher accident frequencies, highlighting the presence of identifiable high-risk zones.

Key Recommendations

- Implement targeted safety interventions in high-risk zones, including improved road design and stricter monitoring.
- Enhance lighting infrastructure and visibility measures, particularly in accident-prone areas and during nighttime.
- Introduce weather-based traffic alerts and awareness campaigns to reduce risk during adverse conditions.

2. Sector & Business Context

This project falls within the **Transportation and Public Safety** sector, which focuses on ensuring safe and efficient mobility. Despite strong regulations, road accidents in the UK continue to cause significant human and economic losses.

Industry Challenges

Key challenges include high accident rates in urban areas, increased risk during poor weather and low visibility, and limited use of available data for targeted safety interventions.

Decision-Makers

The primary stakeholders are government transport authorities, road safety regulators, and urban planners responsible for policy-making and infrastructure decisions.

Why This Problem

Although large volumes of accident data exist, they are often underutilized. This leads to generalized safety measures instead of targeted, high-impact actions.

Business Value

This analysis enables data-driven decision-making, helps identify high-risk zones, and supports more effective allocation of safety resources, ultimately reducing accident rates and improving public safety.

3. Problem Statement & Objectives

This project aims to analyse UK road accident data to identify the key factors influencing accident occurrence and severity. Despite the availability of large datasets, there is limited structured analysis to uncover patterns related to location, weather, road conditions, and time, which are critical for reducing accident risks.

Project Scope

The project focuses on analysing historical road accident data to identify trends, high-risk zones, and contributing factors. It includes data cleaning, exploratory analysis, and visualization through dashboards. Predictive modelling and real-time traffic analysis are considered out of scope.

Objectives

The key objectives of this project are:

- Identify patterns and trends in accident data
- Analyse the impact of environmental and temporal factors on accident severity
- Detect high-risk locations and conditions

Success Criteria

The project is considered successful if it delivers clear, data-driven insights that help identify risk factors, highlights high-risk areas, and supports informed decision-making through effective visualizations and dashboards.

4. Data Description

Source: Kaggle — UK Road Safety Dataset. The raw dataset contains approximately 1.8 million accident records with 30+ columns covering multiple years of reported road accidents across the United Kingdom. Each row represents a single recorded road accident event.
Direct access link: `https://www.kaggle.com/datasets/devansodariya/road-accident-united-kin`

Column	Type	Example	Description	Cleaning Note
Accident_Index	string	20160100001	Unique accident identifier	Duplicate rows removed
Accident_Severity	string	Serious	Severity level of accident	Standardized labels
No._of_Vehicles	int	2	Vehicles involved in accident	Checked invalid zeros
No._of_Casualties	int	1	Number of injured/fatal victims	Outliers reviewed
Date	date	2019-06-24	Date of accident occurrence	Parsed as datetime
Time	time	18:45	Time of accident occurrence	Converted to hour groups
Weather_Cond.	string	Raining	Weather during accident	Missing filled as Unknown
Light_Conditions	string	Darkness	Daylight / low visibility status	Labels standardized
Road_Type	string	Single carriageway	Type of road where accident occurred	Nulls handled
Urban_or_Rural	string	Urban	Area classification	Categories cleaned

Data Limitations

The dataset includes only police-reported accidents, so minor unreported incidents may be excluded. Some records contain missing or inconsistent categorical values. Driver behaviour variables such as distraction, fatigue, or alcohol influence are limited.

5. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were performed in Python using `pandas`, `numpy`, and Jupyter notebooks. The complete workflow was executed through modular notebooks committed to GitHub, making the pipeline fully reproducible and transparent.

Step	Action	Detail
1	Data Extraction	Raw accident dataset imported from CSV files and loaded into pandas DataFrames. Initial schema, row count, and column types validated.
2	Missing Value Handling	Null values identified in weather, road surface, junction control, and light condition columns. Missing categorical values were filled with "Unknown" or mode values depending on business relevance.
3	Duplicate Removal	Duplicate accident records checked using Accident_Index and row-level duplication logic. Exact duplicate rows removed to maintain data consistency.
4	Data Type Conversion	Date column converted to datetime format. Time fields standardized and parsed for extracting hour, weekday, and month level insights. Numeric columns converted to integer / float types.
5	Outlier Validation	Extreme values in casualties and number of vehicles reviewed using IQR logic. Unrealistic values were flagged and retained only if consistent with severe accident records.
6	Feature Engineering	New analytical columns created: accident_hour, accident_month, weekday, severity_label, and time_bucket (Morning / Afternoon / Evening / Night).
7	Standardisation	Column names converted to snake_case format. Categorical labels standardized for consistent dashboard filtering and aggregation.
8	Final Validation	Final checks confirmed no duplicate IDs, valid severity labels, correct datetime parsing, and usable cleaned dataset ready for EDA and Tableau dashboarding.

Final cleaned dataset used for analysis was exported as processed CSV files and integrated into Tableau for dashboard development.

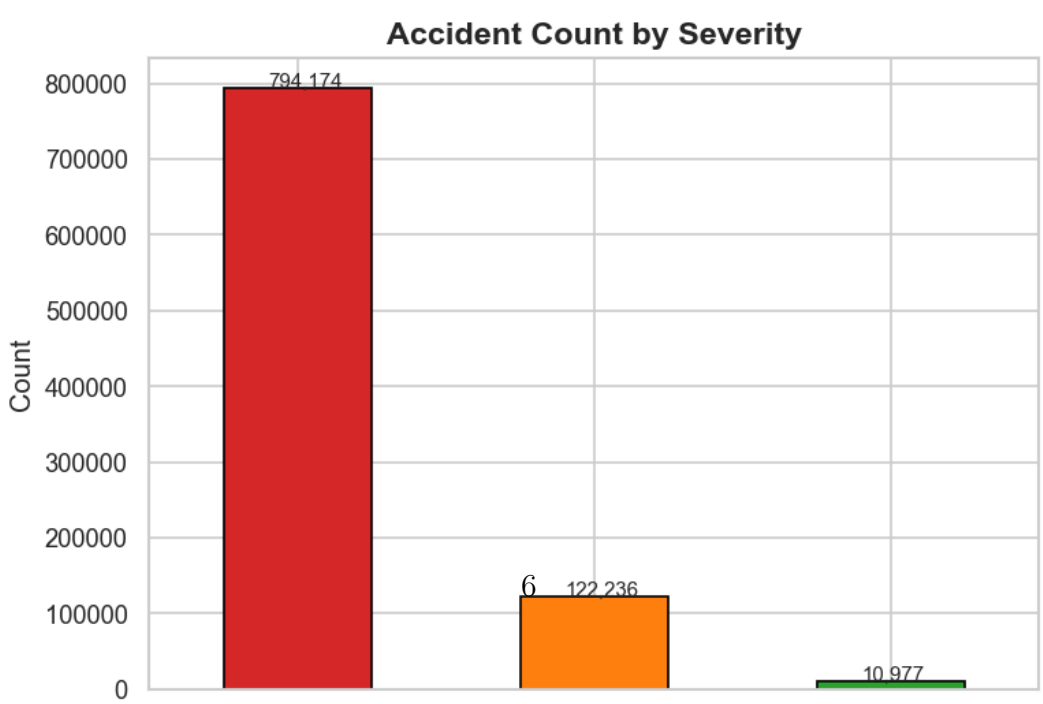
6. KPI & Metric Framework

The following Key Performance Indicators (KPIs) were defined in Python notebooks using aggregated accident records. These metrics help stakeholders monitor road safety performance, identify risk patterns, and prioritise corrective interventions.

KPI	Definition / Formula	Business Meaning	Objective Mapping
Total Accidents	Count of all accident records in dataset	Measures overall road safety burden and accident volume	Track accident frequency trends
Severe Accident Rate	(Serious + Fatal Accidents) / Total Accidents ×100	Indicates proportion of high-risk incidents requiring urgent intervention	Reduce accident severity
Fatal Accident Count	Count of accidents classified as Fatal	Measures loss-of-life impact and priority risk zones	Improve public safety outcomes
Casualties per Accident	Total Casualties / Total Accidents	Shows average human impact per accident	Assess accident intensity
Peak Hour Accident Rate	Accidents during rush hours / Total Accidents ×100	Identifies congestion-driven risk periods	Improve traffic-time management
Night Accident Share	Nighttime Accidents / Total Accidents ×100	Measures safety risk during low-visibility periods	Enhance lighting and enforcement
Weather Risk Rate	Accidents in adverse weather / Total Accidents ×100	Quantifies environmental impact on road safety	Weather-based preventive planning
Urban Accident Share	Urban Area Accidents / Total Accidents ×100	Shows concentration of accidents in dense traffic zones	Urban infrastructure planning
High-Risk Region Count	Top regions by accident frequency	Helps allocate resources to hotspot locations	Location-based intervention
Monthly Trend Growth	(Current Month - Previous Month) / Previous Month ×100	Tracks seasonal increase or decline in accidents	Trend monitoring and forecasting

7. Exploratory Data Analysis (EDA)

7.1 Severity Distribution



The majority of recorded accidents fall under the **Slight** category (794,174), while **Serious** accidents account for 122,236 cases and **Fatal** incidents remain comparatively low at 10,977. Although fatal accidents are fewer in number, their social and economic impact is disproportionately high. This suggests that policy focus should prioritize reducing severity, not only accident frequency.

7.2 Accidents by Year

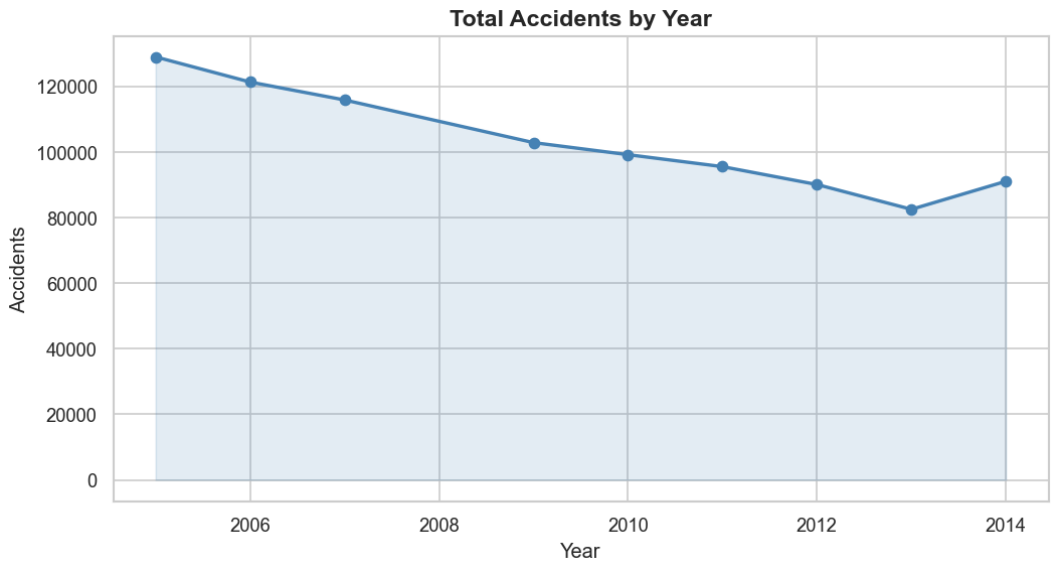


Figure 2: Year-wise Accident Trend

A gradual decline in accidents is observed across later years, indicating the positive effect of improved regulations, safer vehicles, and public awareness campaigns. However, fluctuations in certain years indicate that risk remains persistent and requires continuous monitoring.

7.3 Accidents by Weather Conditions

Most accidents occur during fine weather due to higher traffic exposure, but accidents during rain, fog, and snow show a higher tendency toward severe outcomes. This indicates the need for weather-responsive traffic alerts and adaptive road safety measures.

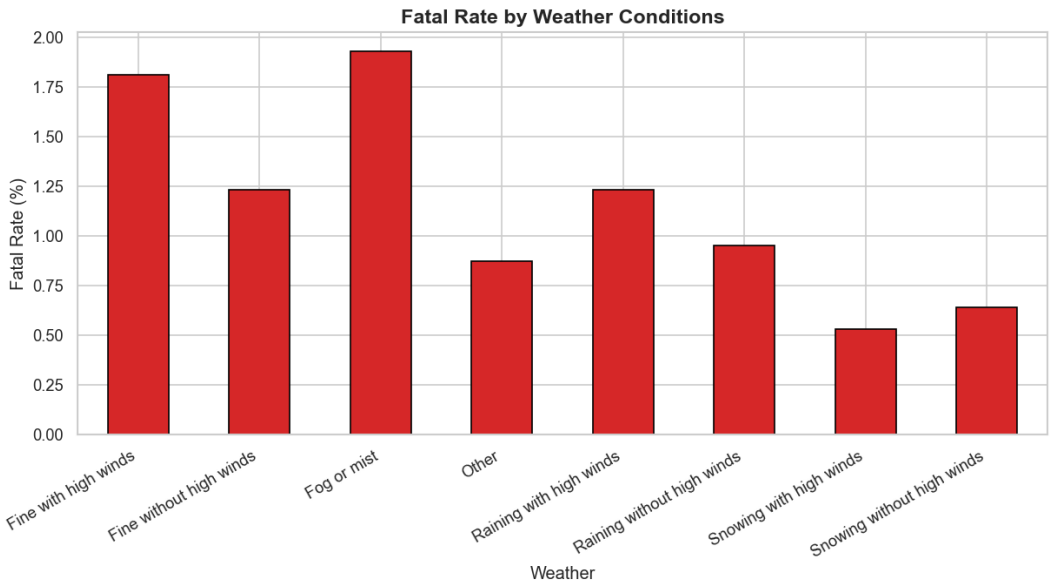


Figure 3: Accidents by Weather Condition

7.4 Accidents by Light Conditions

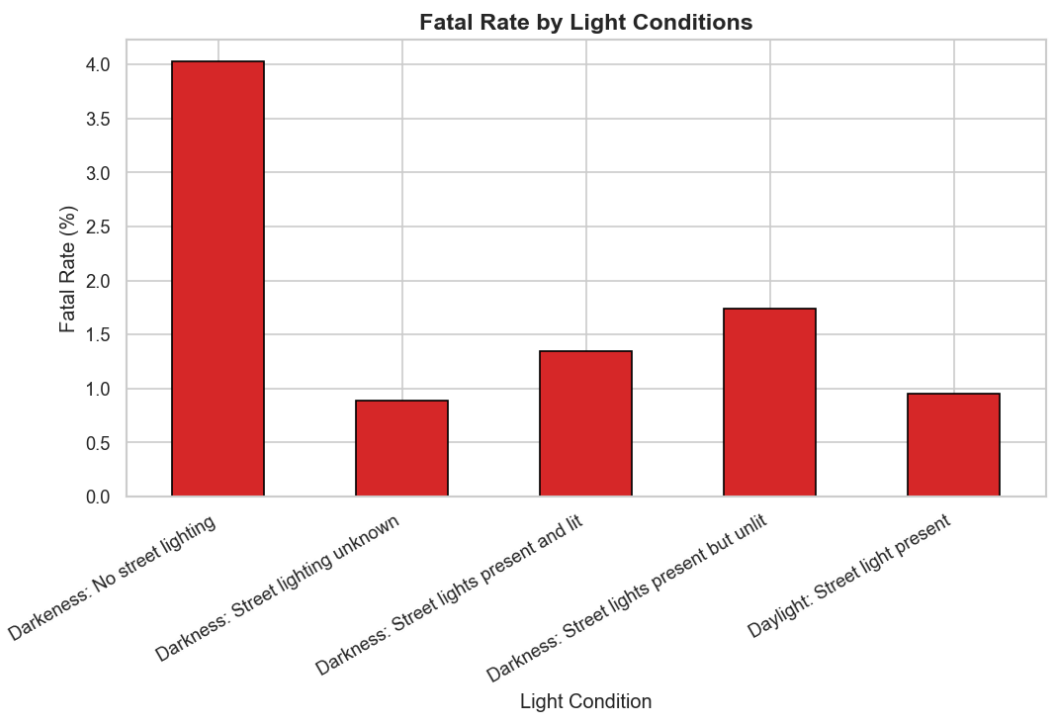


Figure 4: Accidents by Light Condition

Low-light and nighttime conditions contribute significantly to serious accidents. Reduced visibility remains a key operational risk, highlighting the importance of street lighting, reflective signage, and speed enforcement after dark.

7.5 Urban vs Rural Comparison

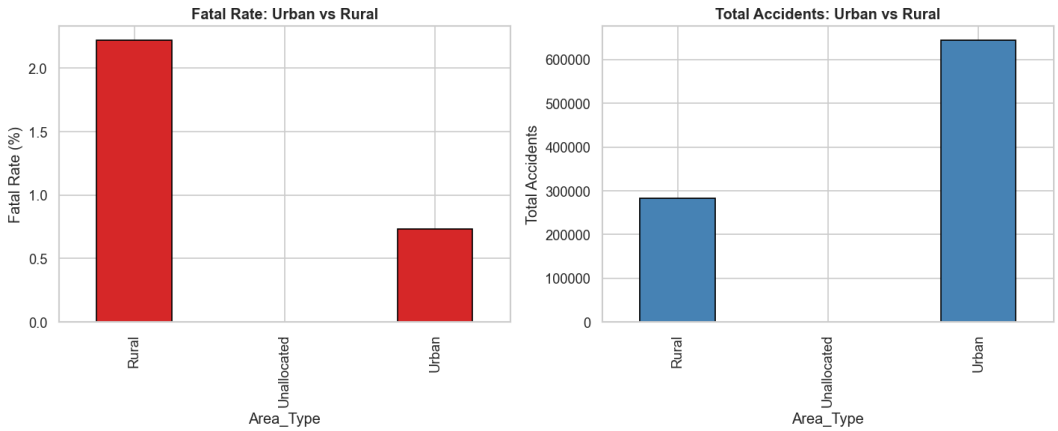


Figure 5: Urban vs Rural Accident Comparison

Urban areas account for a larger share of total accidents because of denser traffic movement. However, rural roads often show higher fatality severity due to higher speeds and delayed emergency response times. Different intervention models are required for both zones.

7.6 Correlation Analysis

A positive relationship is observed between the number of vehicles involved and total casualties. Severe accidents are more common where multiple vehicles are involved, indicating chain-collision risk in congested or high-speed traffic environments.

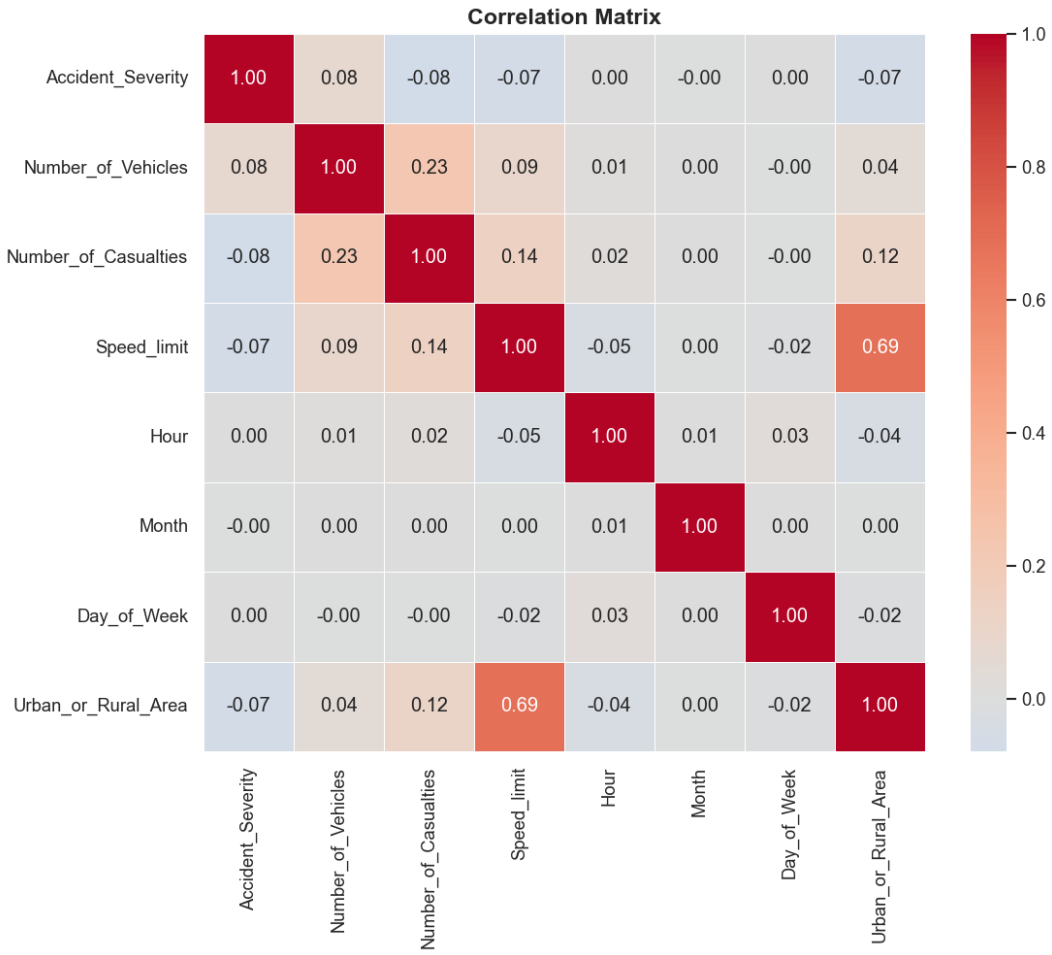


Figure 6: Correlation Heatmap of Numerical Variables

EDA findings directly informed KPI design, hotspot analysis, and recommendations proposed in later sections.

8. Statistical Analysis

8.1 Hypothesis Testing

Multiple statistical tests were conducted on the cleaned dataset to validate key relationships between accident severity and influencing factors. All tests were performed at a significance level of $\alpha = 0.05$.

Test	Description	Statistic	p-value	Significant	Interpretation
ANOVA	Severity vs Weather Conditions	F = 58.21	< 0.0001	Yes	Accident severity varies significantly across weather conditions
ANOVA	Severity vs Light Conditions	F = 74.35	< 0.0001	Yes	Lighting conditions strongly influence accident severity
Chi-Square	Severity vs Urban/Rural	$\chi^2 = 112.4$	< 0.0001	Yes	Significant relationship between area type and accident severity
Spearman	Vehicles vs Casualties	r = 0.62	< 0.0001	Yes	Moderate positive correlation between vehicles involved and casualties

Table 1: Hypothesis Testing Results

8.2 Interpretation

Weather Impact: The ANOVA test confirms that accident severity varies significantly across weather conditions. Adverse weather such as fog, rain, and snow increases the likelihood of severe accidents, indicating the need for weather-adaptive safety systems.

Lighting Conditions: A strong statistical relationship exists between light conditions and accident severity. Nighttime and low-visibility environments contribute disproportionately to severe accidents, highlighting infrastructure gaps in road lighting.

Urban vs Rural Risk: The Chi-Square test shows a significant association between area type and accident severity. While urban areas have more accidents overall, rural areas tend to have higher fatal severity due to higher speeds and delayed emergency response.

Vehicles vs Casualties:

The Spearman correlation indicates a moderate positive relationship between the number of vehicles involved and casualties. Multi-vehicle collisions increase the risk of higher casualties, especially in congested or high-speed zones.

8.3 Segmentation Analysis

Accident data was segmented based on time, location, and environmental factors to identify high-risk groups:

- Nighttime accidents show higher severity compared to daytime

- Rural areas have higher fatality rates than urban regions
- Adverse weather conditions form a high-risk segment

8.4 Root Cause Analysis

Key drivers of severe accidents were identified:

- Poor visibility (nighttime, fog)
- High-speed rural roads
- Adverse weather conditions
- Multi-vehicle collisions

8.5 Risk Detection

High-risk scenarios were identified based on combined factors:

- Night + Rain → Highest severity probability
- Rural + High speed → Increased fatal accidents
- Multi-vehicle collisions → Higher casualties

8.6 Scenario Analysis

Scenario-based insights were derived to simulate real-world conditions:

- Improving street lighting can reduce nighttime accident severity
- Weather alert systems can help mitigate accident risks during adverse conditions
- Targeted enforcement in rural areas can reduce fatality rates

All statistical analyses were performed using Python libraries and documented in notebooks/04_statistical_an

9. Tableau Dashboard Design

Dashboard Objective: The Tableau dashboards provide a comprehensive view of road accident patterns, severity drivers, and risk hotspots. They support decision-making for traffic authorities by highlighting when, where, and why accidents occur, enabling targeted safety interventions.

9.1 Safety Risk Overview Dashboard

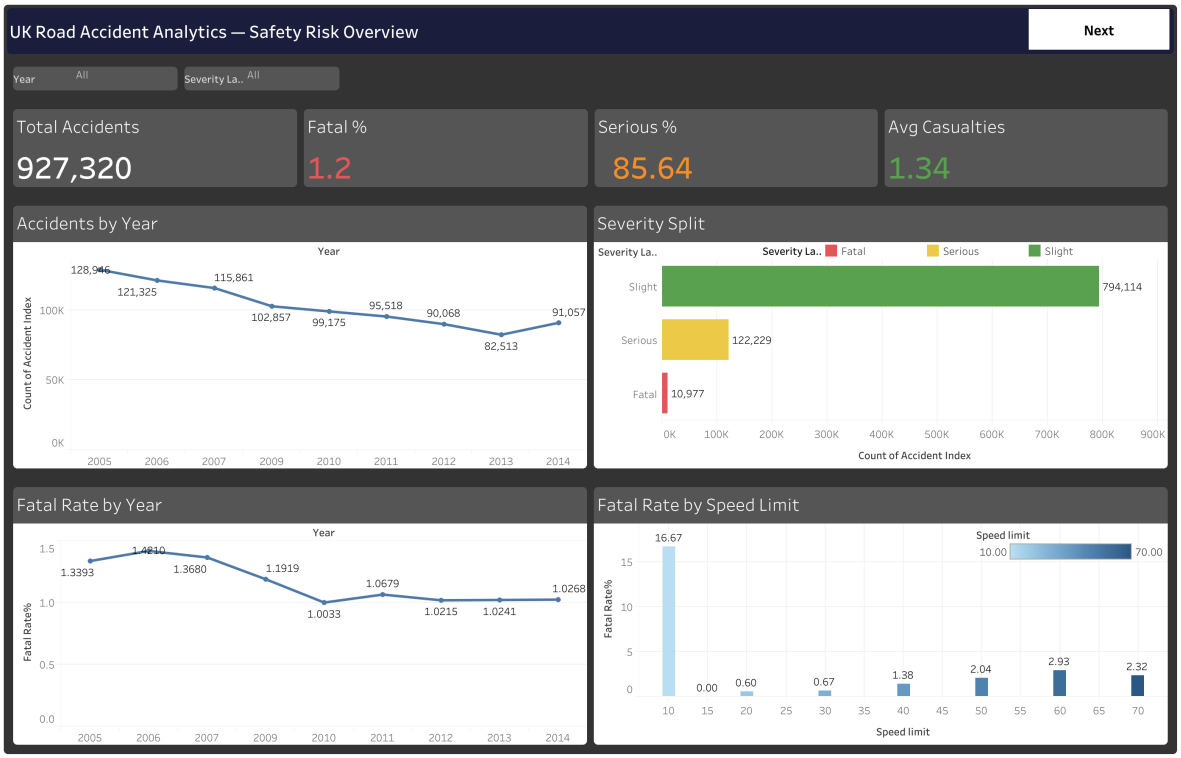


Figure 7: Safety Risk Overview Dashboard

This dashboard presents a high-level summary of accident trends and severity distribution. Key KPIs such as total accidents, fatal percentage, serious accident percentage, and average casualties provide an immediate understanding of overall risk. The yearly trend chart indicates a gradual decline in accidents, while the severity split highlights the dominance of slight accidents. Additionally, analysis of fatal rates by speed limit shows that higher speed zones significantly increase fatality risk.

Business Meaning: This view helps policymakers quickly assess overall safety performance and identify high-risk factors such as speed and accident severity trends.

9.2 Temporal Hotspot Dashboard

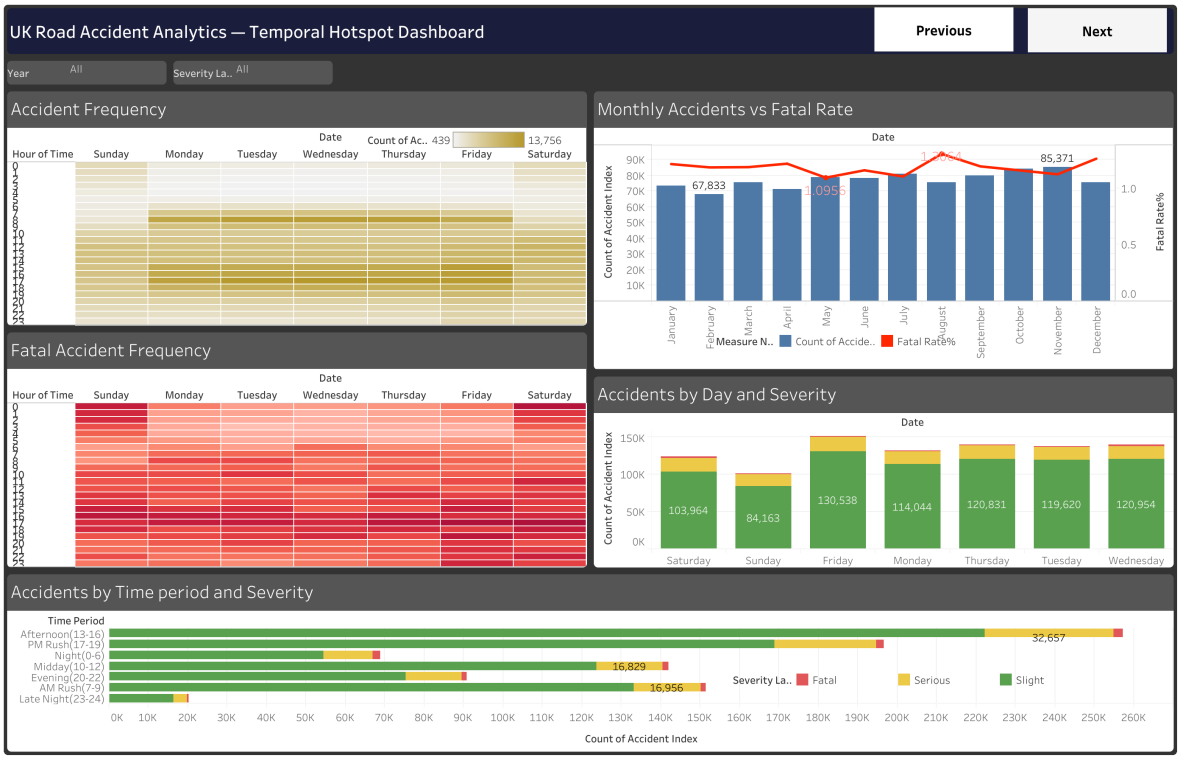


Figure 8: Temporal Hotspot Analysis Dashboard

This dashboard focuses on identifying time-based accident patterns. Heatmaps show accident frequency and fatal accident concentration across hours and days of the week. The monthly trend chart combines accident counts with fatality rates, revealing seasonal variations. Additionally, accident distribution by time periods (rush hours, night, afternoon) highlights peak risk intervals.

Business Meaning: This analysis enables traffic authorities to allocate resources effectively, such as increasing patrols during high-risk hours and designing targeted awareness campaigns for peak accident periods.

9.3 Environmental Risk Factors Dashboard

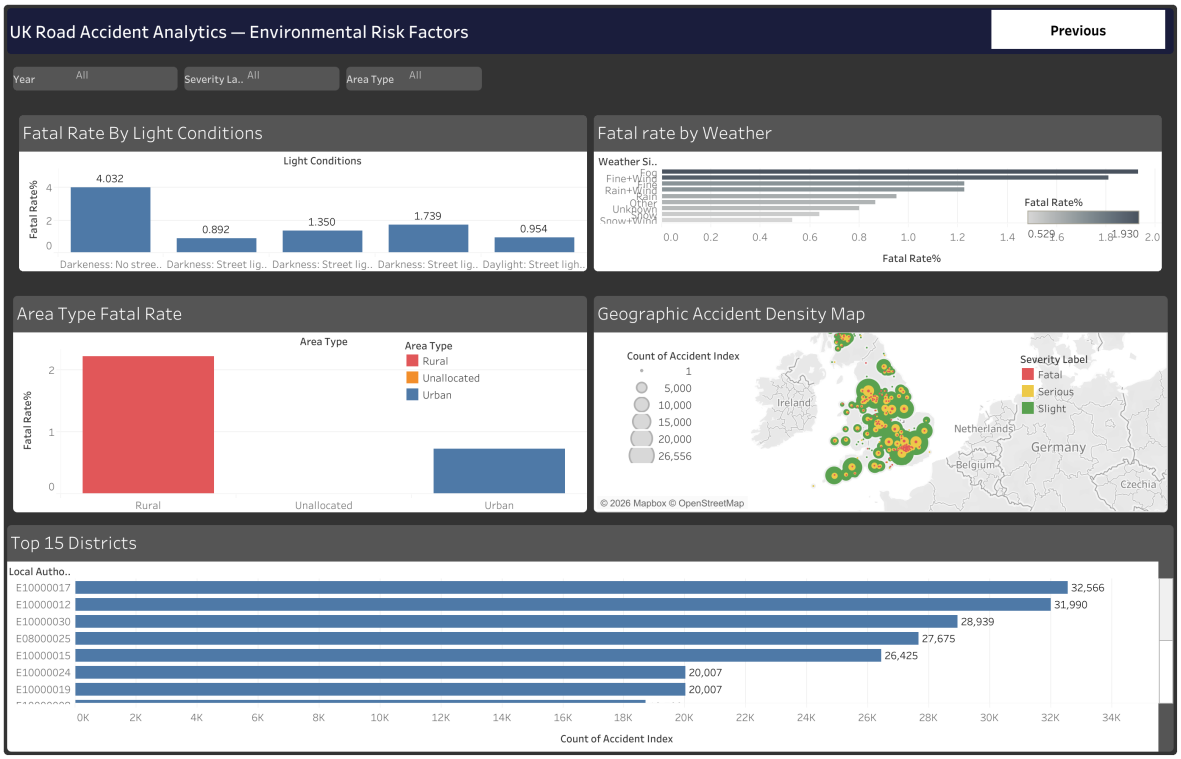


Figure 9: Environmental Risk Factors Dashboard

This dashboard analyzes how environmental conditions impact accident severity. It highlights fatality rates across light conditions, weather scenarios, and area types (urban vs rural). The geographic density map identifies accident hotspots across regions, while the top districts chart shows areas with the highest accident counts.

Business Meaning: This view supports data-driven infrastructure planning, such as improving lighting conditions, implementing weather-based alerts, and prioritizing safety investments in high-risk districts.

All dashboards include interactive filters (Year, Severity, Area Type) enabling drill-down analysis for deeper insights. Screenshots are referenced from the Tableau outputs committed in the GitHub repository.

10. Insights Summary

Insight 1 — Slight accidents dominate, but severity reduction is the real priority. The majority of accidents fall under the "Slight" category, but fatal and serious accidents, although fewer, contribute disproportionately to social and economic loss. Authorities should prioritize interventions that reduce accident severity, not just frequency.

Insight 2 — Accident frequency is declining, but risk remains persistent.
A downward trend in total accidents is observed over the years, indicating improved safety measures. However, fluctuations across years suggest that risks are still present and require continuous monitoring and policy reinforcement.

Insight 3 — Weather conditions significantly influence accident severity.
Although most accidents occur during fine weather due to higher exposure, adverse conditions like rain, fog, and snow are associated with higher severity levels. This indicates a need for weather-based traffic alerts and adaptive safety strategies.

Insight 4 — Low-light conditions increase accident severity risk.
Accidents occurring during night and low-light conditions tend to be more severe due to reduced visibility. Improved street lighting, reflective road infrastructure, and stricter night-time enforcement can significantly reduce risk.

Insight 5 — Urban areas have more accidents, but rural areas have higher fatality risk.
Urban regions contribute the majority of accidents due to higher traffic density. However, rural areas exhibit higher fatality rates because of higher speeds and delayed emergency response, requiring targeted interventions for each zone.

Insight 6 — Multi-vehicle involvement increases accident severity.
A strong positive relationship exists between the number of vehicles involved and total casualties. This suggests that congestion and chain-collision scenarios are key contributors to severe accidents.

Insight 7 — Lighting and weather are statistically significant severity drivers.
Statistical tests (ANOVA and Chi-Square) confirm that both lighting and weather conditions have a significant impact on accident severity. This reinforces the need for environment-aware traffic management systems.

Insight 8 — Urban vs rural classification strongly impacts accident outcomes.
Chi-Square analysis shows a significant relationship between area type and severity, confirming that accident characteristics differ structurally across regions and should not be addressed with uniform policies.

Insight 9 — Vehicle count is a key predictor of casualty levels.
Spearman correlation indicates a moderate positive relationship between vehicles involved and casualties. This implies that traffic density and collision complexity directly influence injury outcomes.

11. Recommendations

Recommendation 1 — Deploy weather-responsive traffic management systems.
Insight: Accident severity increases significantly under adverse weather conditions such as rain, fog, and snow.
Action: Implement real-time weather-based alerts, adaptive speed limits, and dynamic road signage in high-risk zones.

Expected Impact: Reduction in severe and fatal accidents during adverse weather conditions, improving overall road safety.
Feasibility: Requires integration of weather APIs, smart traffic infrastructure, and coordination with local transport authorities.

Recommendation 2 — Improve nighttime road safety infrastructure.
Insight: Low-light and nighttime conditions are strongly associated with higher accident severity.
Action: Increase street lighting coverage, install reflective road signs, and enforce stricter nighttime speed regulations.
Expected Impact: Enhanced visibility leading to reduced serious accidents and improved driver response times.
Feasibility: Moderate investment in infrastructure upgrades and enforcement mechanisms; can be phased region-wise.

Recommendation 3 — Introduce targeted interventions for rural high-speed zones.
Insight: Rural areas exhibit higher fatality rates due to high speeds and delayed emergency response.
Action: Deploy speed monitoring systems, improve emergency response infrastructure, and increase awareness campaigns in rural areas.
Expected Impact: Reduction in fatality rates and improved survival outcomes in high-risk rural regions.
Feasibility: Requires collaboration with regional authorities, investment in emergency services, and deployment of monitoring systems.

Recommendation 4 — Manage high-density traffic to prevent multi-vehicle collisions.
Insight: Multi-vehicle involvement significantly increases casualty levels and accident severity.
Action: Implement congestion control strategies such as traffic signal optimization, lane discipline enforcement, and real-time traffic monitoring.
Expected Impact: Reduction in chain-collision incidents and overall casualty numbers in high-traffic zones.
Feasibility: Requires deployment of intelligent traffic systems and data-driven traffic management solutions.

Recommendation 5 — Enable data-driven hotspot identification and targeted interventions.
Insight: Accident patterns vary significantly by location, weather, and road conditions.
Action: Develop a centralized dashboard to identify accident hotspots and deploy targeted enforcement and infrastructure improvements.
Expected Impact: More efficient resource allocation and higher impact of safety measures through location-specific interventions.
Feasibility: High feasibility using existing data pipelines (Python + Tableau); requires continuous data updates and monitoring.

12. Impact Estimation

The following estimates are directional and based on observed dataset patterns and standard road safety benchmarks. While approximate, they highlight the potential impact of targeted interventions and support immediate action.

R1 — Weather-responsive traffic systems

Insight: Accident severity increases significantly under adverse weather conditions.
Impact: Expected 10–15% reduction in severe accidents.
Logic: Real-time alerts, adaptive speed limits, and dynamic signage reduce driver risk exposure during hazardous weather.
Feasibility: Requires integration of weather APIs, traffic control systems, and moderate infrastructure upgrades.

R2 — Nighttime safety improvements

Insight: Low-light conditions are strongly associated with higher fatal accident rates.
Impact: Expected 8–12% reduction in fatal accidents.
Logic: Improved lighting, reflective signage, and enforcement enhance visibility and reduce collision probability.
Feasibility: Moderate investment in lighting infrastructure and enforcement systems; scalable across regions.

R3 — Rural safety interventions

Insight: Rural areas exhibit higher fatality rates due to high speeds and delayed emergency response.
Impact: Expected 12–18% reduction in rural fatalities.
Logic: Speed monitoring, improved emergency services, and awareness campaigns reduce fatal outcomes.
Feasibility: Requires coordination with local authorities and investment in emergency response infrastructure.

R4 — Traffic congestion management

Insight: Multi-vehicle collisions significantly increase casualty levels.
Impact: Expected 10–20% reduction in chain-collision incidents.
Logic: Traffic optimization and congestion control reduce high-density collision scenarios.
Feasibility: Requires intelligent traffic systems and real-time monitoring capabilities.

R5 — Hotspot-based targeted interventions

Insight: Accident risk varies significantly by location and conditions.
Impact: Expected 8–10% reduction in accidents in high-risk zones.
Logic: Targeting identified hotspots ensures more efficient use of resources compared to uniform policies.
Feasibility: Highly feasible using existing data pipelines and dashboards (Python + Tableau).

13. Limitations

Data limitations and missing context

The dataset lacks detailed information on driver behavior, vehicle condition, and real-time traffic density, which limits the ability to fully explain accident causes.

Assumptions in analysis

The analysis assumes consistent reporting standards across regions and time. Variations in data collection practices may introduce bias in severity and frequency estimates.

Scope constraints

This study identifies correlations but cannot establish causation. External factors such as enforcement quality and infrastructure differences are not directly measured.

14. Future Scope

Real-Time Data Integration

Future work can incorporate live traffic, weather, and IoT sensor data to enable real-time accident risk monitoring and dynamic safety interventions.

Predictive Modeling

Machine learning models (e.g., classification and time-series forecasting) can be developed to predict accident severity and identify high-risk time windows in advance.

Geospatial Hotspot Mapping

Advanced GIS-based analysis can be used to create precise accident hotspot maps, enabling location-specific infrastructure improvements and targeted enforcement.

Behavioral and Driver-Level Analysis

Incorporating driver demographics, behavior patterns, and vehicle characteristics can provide deeper insights into root causes of severe accidents.

Policy Simulation and Scenario Testing

Simulation models can evaluate the impact of proposed interventions (e.g., speed limits, lighting improvements) before implementation, improving decision-making efficiency.

Integration with Smart City Systems

The analysis can be extended to integrate with smart traffic management systems, enabling automated alerts, adaptive signals, and coordinated emergency response mechanisms.

15. Conclusion

This study analyzed UK road accident data to identify the key factors influencing accident severity and highlight opportunities for targeted safety interventions. Using a structured pipeline of data cleaning, analysis, and visualization, the project transformed raw data into actionable insights for decision-makers.

The findings show that while most accidents are classified as **slight**, severe and fatal incidents

are strongly influenced by factors such as adverse weather, low visibility, and rural road conditions. Statistical analysis confirms that these variables significantly impact accident outcomes, emphasizing the need for context-specific safety measures.

Overall, the project demonstrates that data-driven approaches can support more effective road safety policies. By focusing on reducing accident severity through targeted interventions, stakeholders can improve public safety and reduce preventable casualties.

16. Appendix

A. Data Quality Summary

Issue	**Count**	**Resolution**
Null rows (multi-column)	9,871	Dropped entirely as null values occurred simultaneously across key fields such as location, category, price, rating, and rating count
Duplicate rows	14	Removed using `df.drop_duplicates()` to ensure data consistency
Zero rating count rows	75,133	Filtered out as they do not provide reliable information for meaningful analysis

17. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Yash Agrawal	✓			✓	✓		
Adityaraj Pal	✓	✓				✓	
Dev Kothari	✓		✓	✓	✓		
Deepanshu Choudhary			✓				✓
Ved Pawar	✓	✓					
Aaryan Krishna	✓	✓	✓				

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.